

Investigating the Causes of Imbalance in Buses and Practical Solutions

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Abstract

In daily life, standing passengers on buses often struggle to maintain their balance, especially during rapid acceleration, deceleration, or turns. These challenges highlight the difficulties of staying upright in a dynamic and often unpredictable environment. This report investigates the relative importance of sliding motion versus rotational motion in causing loss of balance. The methodology includes four key steps: data collection, function fitting, application of physical principles, and numerical calculations. Through this analysis, we conclude that rotational motion plays a more significant role in causing imbalance in buses. The findings of this study provide practical suggestions for passengers, based on torque balance considerations.

1 Introduction

In Macao, passengers on buses often experience difficulty in maintaining their balance, and in some cases, may even fall. This report aims to explore the reasons behind this phenomenon and propose practical solutions. During the literature review, we found that much of the existing research on moving buses focuses on topics such as safety, transportation, and pure data analysis, but few studies address the problem from a physics-based perspective. In this report, we will present an overview of the analysis methods, data and function fitting, calculations, and the analysis of balance loss. Additionally, we will discuss practical countermeasures.

2 Analysis Methods Overview

This section outlines the analytical framework for investigating passenger balance and stability in buses, particularly focusing on the reasons why standing passengers may lose balance. The discussion integrates fundamental physics principles, and mathematical techniques to establish a systematic method for analysis.

(1) Newton's Laws of Motion and Initial Considerations

A force, when unopposed, will alter the state of motion of an object, as per Newton's laws of motion. Applying this principle, the forces acting on a standing passenger during bus movement are of central interest. Since directly measuring these forces can be challenging, we turn to Newton's second law:

$$\vec{F} = m \vec{a} = m \frac{d \vec{v}(t)}{dt}$$

where $v(t)$ represents the velocity of the passenger as a function of time t . This provides a pathway to analyze the force upon the passenger through velocity data of bus.

(2) Data Collection

In order to make the final results relatively reliable, we selected two sets of data one after 11 midnight and the other one after 12 noon. At midnight, we measured the Lotus roundabout, and at noon we measured the Shipai Wan Road.

(3) Velocity, Acceleration, and Derived Formulas

Given the difficulty of obtaining direct measurements of force, we analyze the speed of the passenger, which is often available from bus motion data. Mathematical techniques are employed to derive acceleration formulas based on the speed function $v(t)$.

Straight-Line Motion

In typical scenarios, buses primarily travel in a single direction without reversing. The acceleration in this case can be directly obtained as the time derivative of the speed:

$$a_0(t) = \frac{dv(t)}{dt}$$

Circular Motion with Constant Radius

For curved paths, such as turns in a roundabout, the motion can be analyzed in planar polar coordinates. The total acceleration is resolved into two components:

- **Radial Acceleration** ($a_1(t)$): Directed towards the center of the circular path, given by:

$$a_1(t) = \frac{v(t)^2}{radius}$$

- **Transverse Acceleration** ($a_2(t)$): Aligned tangentially, equivalent to $a_0(t)$ formula.

The net acceleration, $a_3(t)$, combines these components using the Pythagorean theorem:

$$a_3(t) = \sqrt{a_1^2 + a_2^2}$$

(4) Force Analysis and Passenger Stability

For a simpler analysis, we choose the bus, non-inertially, as the reference frame. With the accelerations $a_0(t)$ and $a_3(t)$ determined, the corresponding inertial forces on

a passenger of mass m can be calculated as $F_{inertial} = -m a_{bus}$. We will discuss the derivation of the formula in Section Four. These forces serve as the basis for understanding balance issues.

Sliding Motion

A passenger may slide if the external forces (e.g., static friction or normal forces from support structures like poles) are unsuitable to keep the acceleration = 0 in a bus environment. Sliding continues until additional external forces provide the necessary support.

Rotational Motion

Maintaining rotational balance involves ensuring that the net torque on the passenger is zero. Failure to counteract these torques could cause the passenger to tilt too much to the vertical.

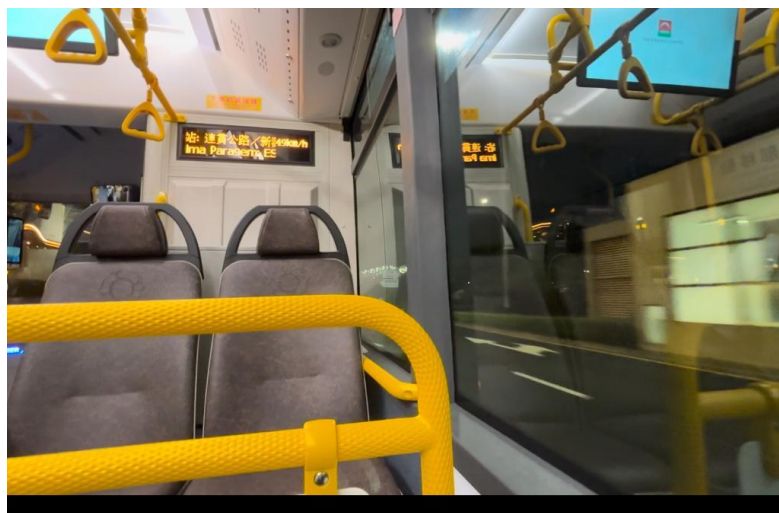
(5) Practical Analysis and Conclusions

Using available data—including bus motion, passenger physical limits, and friction coefficients—we will identify the primary causes of instability. Both sliding and rotational motions are considered as potential contributors, with a focus on quantifying the limits of human stability under typical bus conditions.

3 Measurement of Speed, Function Fitting

For the speed data collection, we selected typical road conditions, including turnings, roundabout, and straight-line acceleration and deceleration. An HD camera was used to record the speed display on a bus in Macao. **Figure 1** shows an example of the data recording process.

In total, we collected two sets of data. The first set pertains to the circular road condition, and the second set corresponds to the turning and straight-line conditions.



In the following sections, we will use MATLAB to fit some different $v(t)$ function expressions to these datasets.

The first data set was collected at the Lotus roundabout. The segment of the road used for data collection is shown in **Figure 2**.

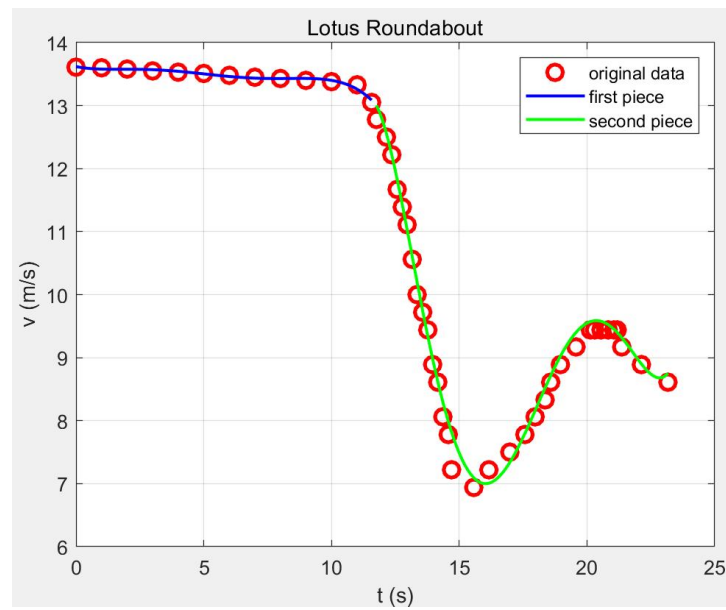
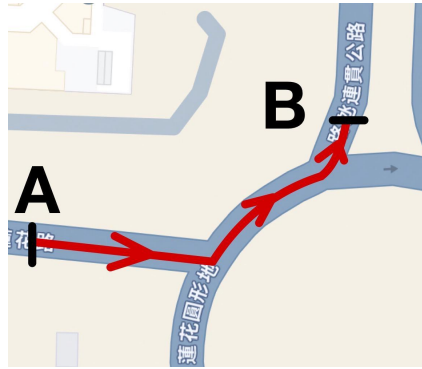


Figure 3 displays the original data points, marked with red circles. We applied two piecewise fitting functions to this data, as a single polynomial fit did not accurately capture the pattern of the speed data.

The corresponding functions are denoted by f_1 , and f_2 for $t \in [0, 11.75]$ and

$$f_1 = -\frac{2876765661083815}{36893488147419103232} \cdot t^5 + \frac{4534394594956829}{2305843009213693952} \cdot t^4$$

$$-\frac{2452462513149079}{144115188075855872} \cdot t^3 + \frac{8301000808451963}{144115188075855872} \cdot t^2$$

$$-\frac{5970382441135957}{72057594037927936} \cdot t + \frac{7667030127498041}{562949953421312}$$

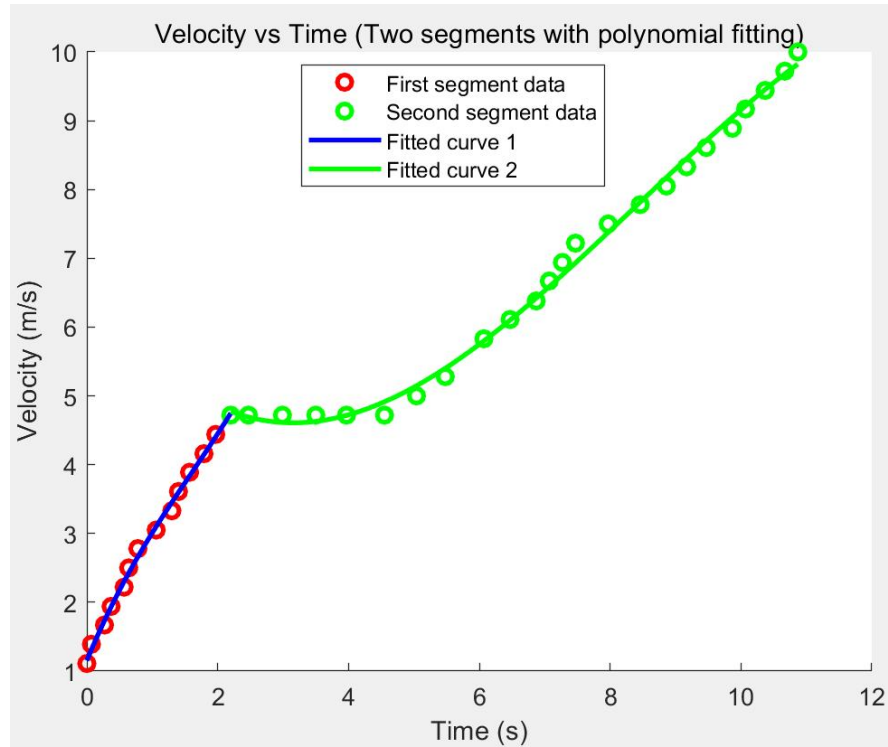
$$f_2 = \frac{2902488795913839}{2305843009213693952} \cdot t^5 - \frac{1001884068544077}{9007199254740992} \cdot t^4$$

$$+ \frac{8675560934811535}{2251799813685248} \cdot t^3 - \frac{573589214890645}{8796093022208} \cdot t^2$$

$$+ \frac{2364993669833955}{4398046511104} \cdot t - \frac{7538867925767329}{4398046511104}$$

$t \in [11.75, 23]$ respectively. Here, the units of speed is m/s.





The second data set was collected from a turning at the intersection of Shipaiwan Road and a smaller connecting road. The road segment used for data collection is shown in [Figure 4](#).

[Figure 5](#) shows the original data points (marked as circles). As with the first data set, we used two piecewise fitting functions to model the speed data, since a single polynomial fit did not provide an adequate representation.

We get the corresponding $v(t)$ expressions denoted by g_1 and g_2 .

$$t \in [0, 2.2], g_1(t) = 0.1163t^3 - 0.5570t^2 + 2.2968t + 1.1532$$

$$t \in [2.2, 10.87], g_2(t) = -0.0109t^3 + 0.2751t^2 - 1.4089t + 6.6585$$

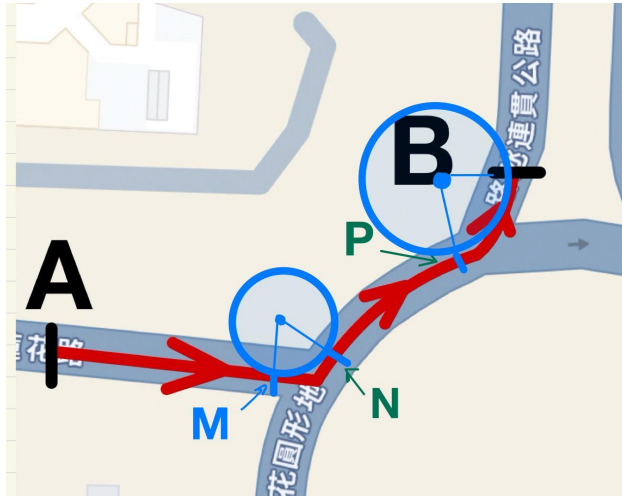
4 Calculations and Analysis for not Maintaining Balance

First of all, we calculate accelerations expressions for each speed derived from data sets.

For the first data set, we divide the path into four segments, as the two figures , **Figure 6 and Figure 7** shown.

By Google Earth , we get following approximated radii data. The left-most radius denoted by r_1 is 20.6 m. The red radius denoted by r_2 is 65 m. The up-most radius denoted by r_3 is 34.3 m. When $t \in [0, 14]$, the bus is in the AM segment. when $t \in [14, 17]$, the bus is in the arc MN. when $t \in [17, 20]$, the bus is in the arc NP. When $t \in [20, 23]$, the bus is in the arc PB. Through calculation methods mentioned in Section 2, we have following results.





$$f_1'(t) = -0.0004 \cdot t^4 + 0.0079 \cdot t^3 - 0.0511 \cdot t^2 + 0.1152 \cdot t - 0.0829$$

$$f_2'(t) = 0.0063 \cdot t^4 - 0.4449 \cdot t^3 + 11.5582 \cdot t^2 - 130.4191 \cdot t + 537.7373$$

$$t \in [0, 11.75], \quad a_0 = f_1',$$

$$t \in [11.75, 14], \quad a_0 = f_2',$$

$$t \in [14, 17], \quad a_1 = [f_2]^2 / r_1$$

$$= 0.048544 \cdot \left(0.0012588 \cdot t^5 - 0.11123 \cdot t^4 + 3.8527 \cdot t^3 - 65.21 \cdot t^2 + 537.74 \cdot t - 1714.1 \right)^2$$

$$a_2 = f_2'$$

$$a_3 = \sqrt{0.0023565 \cdot \left(0.0012588 \cdot t^5 - 0.11123 \cdot t^4 + 3.8527 \cdot t^3 - 65.21 \cdot t^2 + 537.74 \cdot t - 1714.1 \right)^4 + \left(0.0063 \cdot t^4 - 0.4449 \cdot t^3 + 11.558 \cdot t^2 - 130.42 \cdot t + 537.74 \right)^2}$$

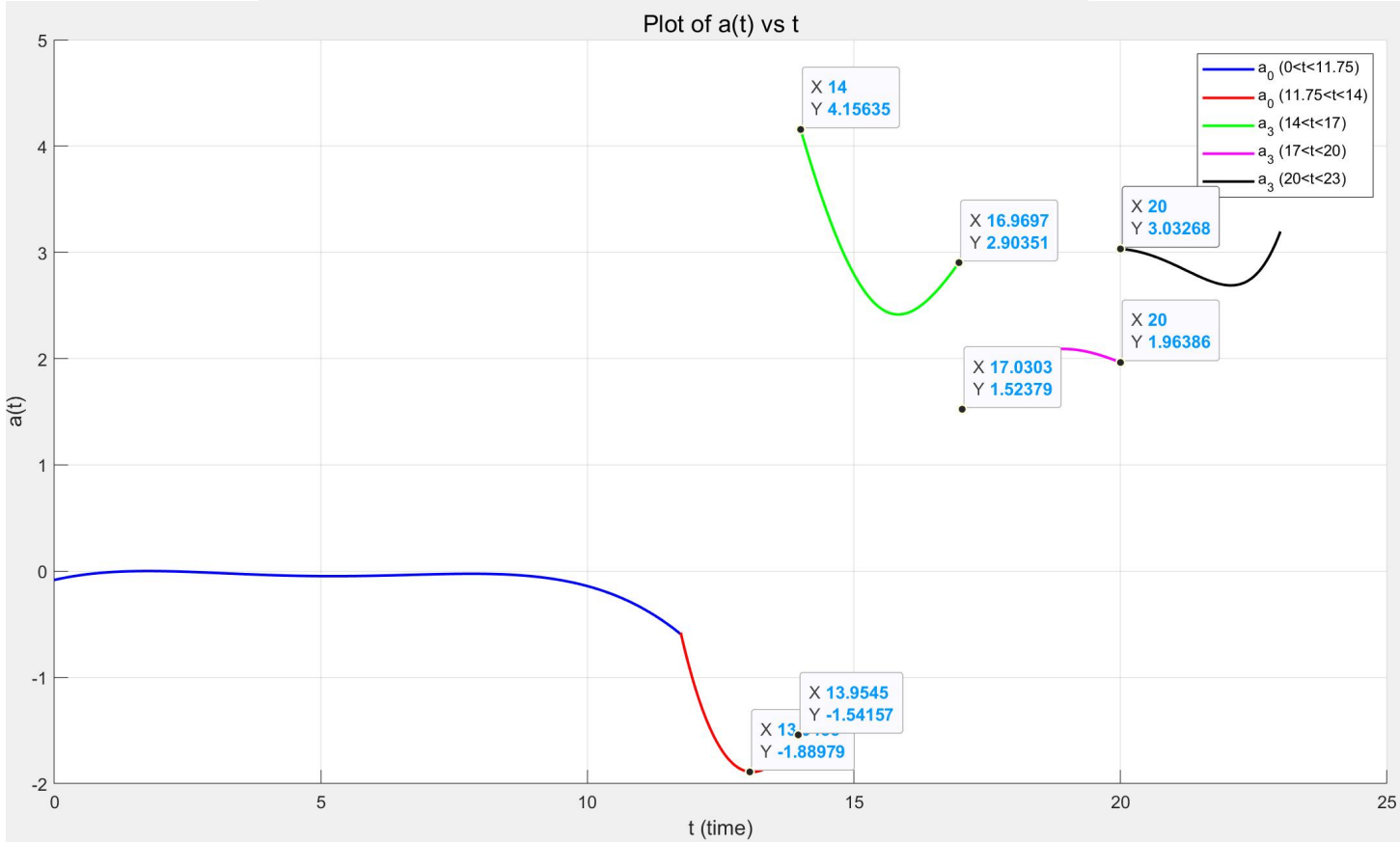
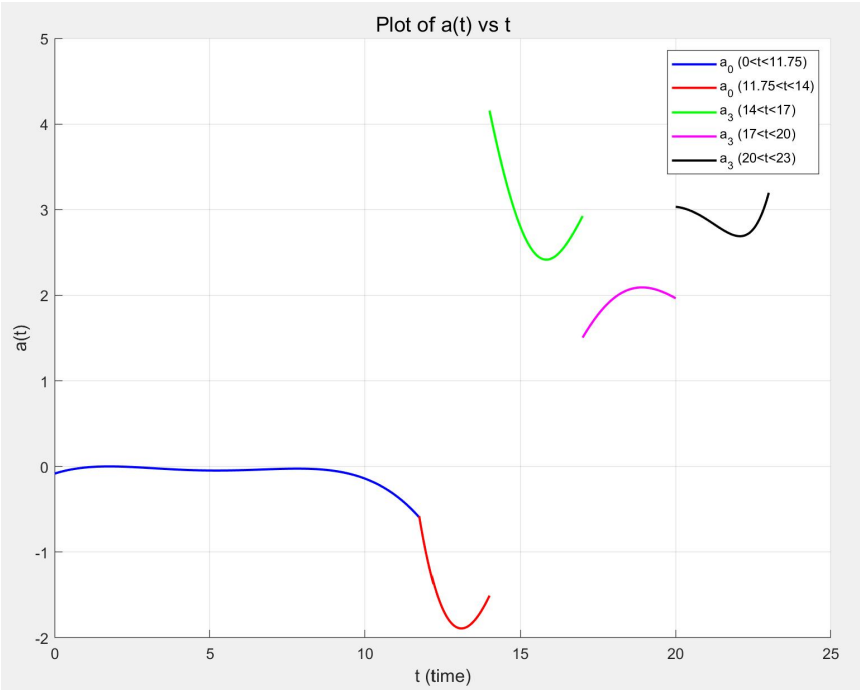
$$t \in [17, 20], \quad a_1 = [f_2]^2 / r_2, \quad a_2 = f_2',$$

$$a_3 = \sqrt{0.0002367 \cdot \left(0.0012588 \cdot t^5 - 0.11123 \cdot t^4 + 3.8527 \cdot t^3 - 65.21 \cdot t^2 + 537.74 \cdot t - 1714.1 \right)^4 + \left(0.0063 \cdot t^4 - 0.4449 \cdot t^3 + 11.558 \cdot t^2 - 130.42 \cdot t + 537.74 \right)^2}$$

$$t \in [20, 23], \quad a_1 = [f_2]^2 / r_3, \quad a_2 = f_2',$$

$$a_3 = \sqrt{0.00084999 \cdot \left(0.0012588 t^5 - 0.11123 t^4 + 3.8527 t^3 - 65.21 t^2 + 537.74 t - 1714.1 \right)^4 + \left(0.0063 t^4 - 0.4449 t^3 + 11.558 t^2 - 130.42 t + 537.74 \right)^2}$$

Through these results, we plot them in one graph with two editions named as [Figure 8](#)



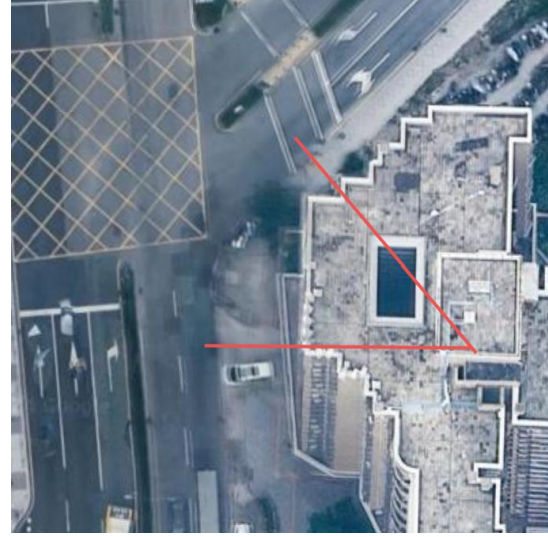
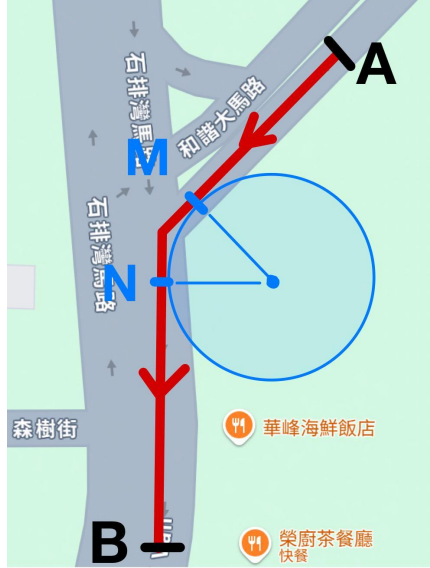
and 9. The difference is that the latter one has discontinuous points information.

Since the centripetal acceleration for each radius occurs suddenly, the net accelerations could suddenly change. i.e. the discontinuous points of the image are reasonable.

In the figure with data labels, we could see the maximum acceleration occurring at $t =$

14s with $a = 4.16 \text{ m/s}^2$ denoted by c_1 .

From the second data set, we divide the path into three segments, as the two figures shown, named as **Figure 10, and 11**. We use the left-hand side figure to illustrate the path condition. In other words, in the left picture the actual length is not concerned.



By Google Earth , we get following approximated radius data. The radius denoted by

r_4 is 25.5m. When $t \in [0, 1]$, the bus is in the AM segment. when $t \in [1, 7]$, the bus is in the arc MN. when $t \in [7, 10.87]$, the bus is in the arc NB. Through calculation methods mentioned in Section 2, we have following results.

$$t \in [0, 1]: [a_0(t) = 0.24t^2 - 1.12t + 2.3]$$

$$t \in [1, 2.2]: [a_{31}(t) = (\sqrt{26010000(12t^2 - 56t + 115)^2 + (12t^3 - 56t^2 + 230t + 115)^4} / 255000)]$$

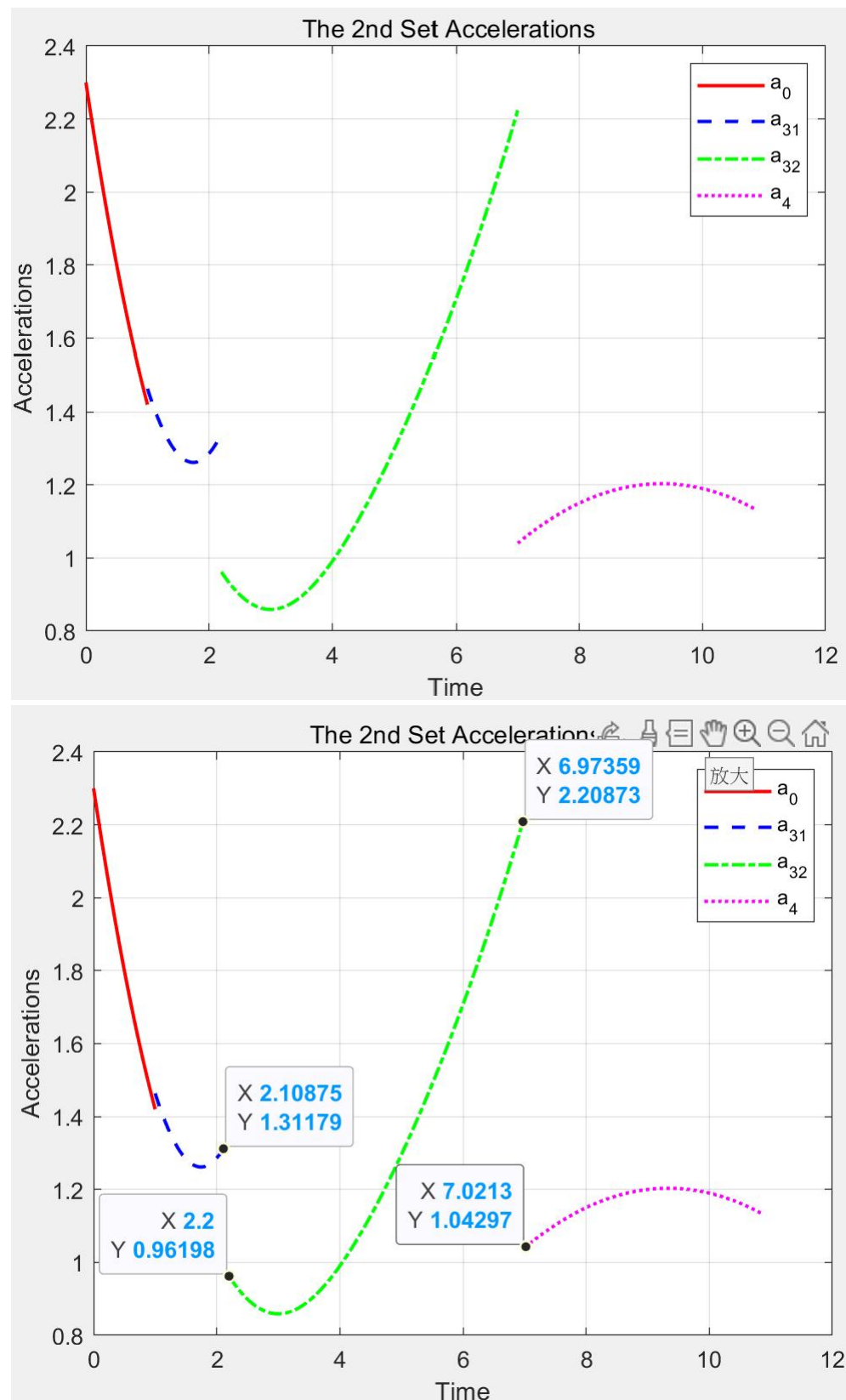
$$t \in [2.2, 7]: [a_{32}(t) = \sqrt{((t^3 - 28t^2 + 141t - 666)^4 + 6502500(3t^2 - 56t + 141)^2) / 255000}]$$

$$t \in [7, 10.87]: [a_4(t) = -0.03t^2 + 0.56t - 1.41]$$

Since the speed expressions are divided into two formulas at $t = 2.2$, we have four expressions for accelerations.

As we plot them in one graph with two figures by **Figure 12 and 13**, we could find that there are four parts in these accelerations, i.e. three discontinuous points. For the first $t = 1s$, since the bus gets into the circular path, we could see that the acceleration gets a little bit bigger. For the second $t = 2.2s$, since the bus transverse acceleration

suddenly declines as we have seen in the Figure 5 in the function fitting section. For the third $t = 7s$, since the bus gets out of the circular path, and the centripetal acceleration suddenly vanishes.



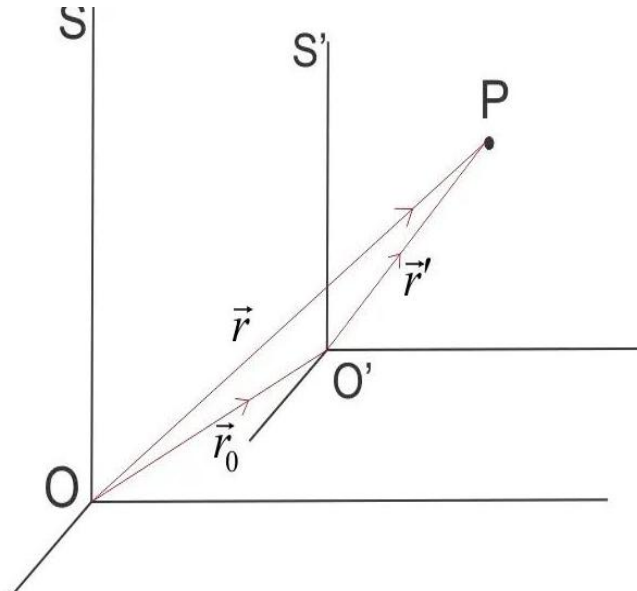
From the second visualized figure, we can see that the biggest acceleration value is

less than the acceleration c_1 in the first data set. For the detailed analysis in this section, we will not use the acceleration of data set 2. In the Section 5, we will use the graph features of the second figure to help us develop some methods to keep balance in the bus.

Now, we will analyze the possible sliding and rotational motion of a passenger. All motion analysis uses the bus as a reference frame, a non-inertial frame. We consider the moving bus as a particle.

We will illustrate a modified Newton's second law suitable for the non-inertial frame.

[1]



We set two reference frames named by S inertial, and S' non-inertial in [Figure 14](#).

$$\vec{r} = \vec{r}' + \vec{r}_{ss'}$$

$$\vec{v} = \vec{v}' + \vec{v}_{ss'}$$

$$\vec{a} = \vec{a}' + \vec{a}_{ss'}$$

Since we want to use $\vec{a}'$. Let

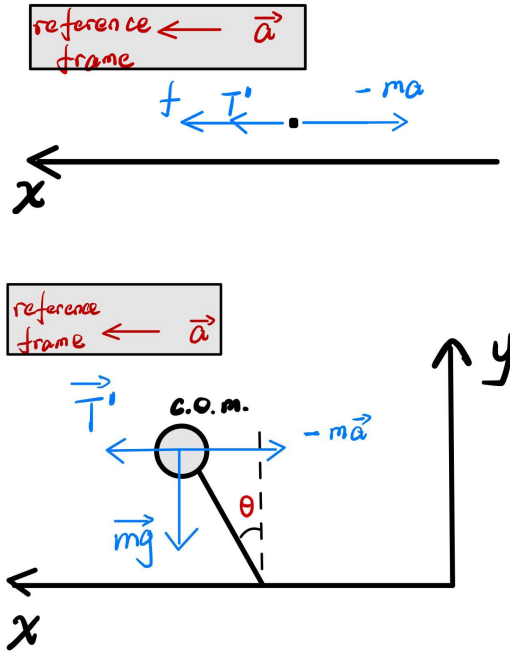
$$\vec{F}' = m\vec{a}' = m\vec{a} - m\vec{a}_{ss'} = \vec{F} + \vec{F}_{inertial}$$

In short, whenever we add inertial force to the net force, we could use the acceleration observed in a non-inertial reference frame like that from an inertial one. We could treat the sliding motion as one dimensional problem, and the rotational motion as two dimensional problem.

Consider the balance and stability conditions of the passenger, we have following two equations. We name the upper one as eq1 and the other as eq2.

$$T' + f - ma = ma' = 0$$

$$(T' \cos \theta + G \sin \theta - m a \cos \theta) L = I (a' / L) = I a' = 0$$



There are two corresponding diagrams, [Figure 15](#) and [Figure 16](#).

The T' is the reaction force of supports such as handrails. The f is the static frictional force between the passenger

and the bus floor. The θ is the little inclined angle between the human body and the vertical line.

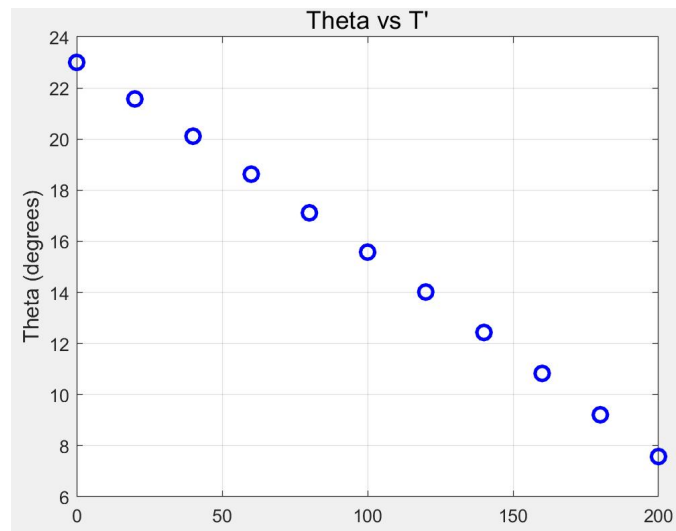
The G , I , L , a' are gravitational force, rotational inertia, distance between center of mass and rotational axis (foot), and angle

Rubber	Cardboard	Clean and Dry	0.5 - 0.8
Rubber	Dry Asphalt	Clean and Dry	0.9

acceleration of the passenger respectively. For simplification of rotational analysis, we think the inertial force ma and T' both acting on the center of mass.

We suppose that the considered passenger is a 20-year-old male with rubber-soled

shoes. Parameters m , g , μ_s are set as 70 kg, 9.8 m/s^2 , 0.7. The [Table 1 \[2\]](#) is



about static frictional coefficient. We set $T' \in [-400N, 400N]$, referring to the

eligibility line for grip strength scores in adults (20-24 years old) from the National Physical Fitness Measurement Standards[3]. We could get that $f \in [-480.2N, 480.2N]$.

For eq1 and eq2, we plug the c_1 of the first data set result into the acceleration a.

$$T' + f = 70 \times (c_1) = 291.2N < f_{max}$$

Since the continuous motion of passenger is difficult to analyze, we choose some discretized points and analyze balance conditions at those points. For eq2, we choose 11 discrete values from 0 to 200 of T' as a description of trends, as shown in [Figure 17](#).

From the result of eq1, we could see that the passenger could keep in stability even only static frictional force supporting. From the figure derived from the result of eq2, we could see that if no T', the passenger need to lean 23 degrees to maintain their balance without turning. The number of 23 degrees is a relatively big angle for a passenger standing in a bus. **As a result, we could conclude** that failure to maintain torque balance is the main reason why passengers cannot maintain balance in a moving bus.

5. Practical Methods for Maintaining Balance

In the previous section, we concluded that rotational motion is the primary cause of imbalance. Based on additional references and the torque analysis, we propose the following practical suggestions to help passengers maintain balance.

(a) Adjusting the Tilt Angle of the Body

By adjusting the body tilt, passengers can maintain torque balance without relying on excessive arm strength. As shown in the [Figure 17](#) in Section 4, we observe that if passengers can automatically adjust their body tilt angle according to the bus's driving conditions, they can achieve balance by relying solely on gravitational torque (see the point of [Figure 17](#) in Section 4, where $T'=0$).

In short, we summarize a practical method as followed. The direction in which the car is accelerating, the passenger should lean in that direction. i.e. When the bus accelerates forward, the passenger should lean forward. When the bus accelerates backward, the passenger should lean backward. When the bus turns left, the passenger should lean leftward. When the bus turns right, the passenger should lean rightward.

(b) Choosing an Optimal Position and Enhancing Reaction and Balance Abilities

According to a study by Darowski[4], maintaining postural stability requires the brain to process complex multisensory information from the task and the surrounding environment. This information is gathered through the body's vestibular, somatosensory, and visual systems. From the acceleration versus time graphs presented in Section 4, we can see that the instantaneous change in acceleration is relatively large. For example, in Data Set 1, the largest acceleration change is 5.7 m/s^2 , and in Data Set 2, it is 1.1 m/s^2 . For a passenger weighing 70 kg, this translates into a force change of 399 N or 77 N.

In order to counteract these changes in force and maintain balance, passengers need to react quickly. If a passenger intends to use the method described in part (a), the key challenge is whether they can adjust their posture rapidly enough to accommodate the bus's acceleration. We recommend two strategies to improve this ability:

1. Stand in a position where you can see the road ahead. This allows the passenger to anticipate the bus's movements and adjust their posture proactively.
2. Strengthen physical fitness, particularly through exercises like running and fast walking, which can enhance balance and reaction time.

6 Conclusion

This report investigates the main cause of imbalance experienced by passengers in moving buses. Our findings indicate that rotational motion plays a more significant role in causing imbalance than sliding motion. The significance of this report lies in its demonstration that balance issues can be addressed by adjusting the tilting angle of the body. However, the study has two main limitations. First, the data collection process was not comprehensive or sufficiently accurate. Second, the supporting force

acting on the passenger's body in real-life scenarios is more complex than the simplified model (T') used in Section 4. In the future, with better data collection techniques and more advanced analytical tools, more detailed and valuable results may be obtained.

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